

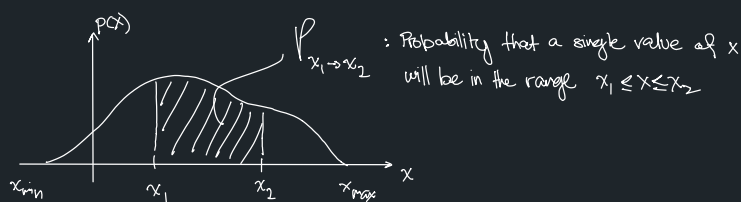
Random variable X

A variable whose value cannot be known in advance

Can only calculate the probability that a random variable will be in a given range.

We treat experimentally measured quantities as random variables

Probability Density Function



$$P_{x_1 \rightarrow x_2} = \int_{x_1}^{x_2} p(x) dx$$

← pdf or density

$$P_{x_{\min} \rightarrow x_{\max}} = 1$$

$$\text{mean value of } x: \mu_x = \int_{x_{\min}}^{x_{\max}} x p(x) dx$$

$$\text{variance of } x: \sigma_x^2 = \int_{x_{\min}}^{x_{\max}} (x - \mu_x)^2 p(x) dx$$

CUMULATIVE DISTRIBUTION FUNCTION (CDF)

$$C(x) = \int_{x_{\min}}^x p(x) dx = P_{x_{\min} \rightarrow x}$$

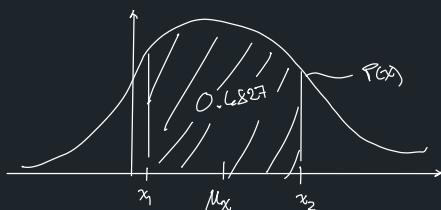
Distributions (PDFs) relevant to ENG 109

Normal (Gaussian)
Student's t
 χ^2 (Chi-square)
Triangular
Uniform

① Normal (Gaussian)

$$p(x) = \frac{1}{\sqrt{2\pi} \sigma_x} \exp \left[-\frac{(x - \mu_x)^2}{2\sigma_x^2} \right]$$

$$x_{\min} = -\infty, x_{\max} = \infty$$



where μ_x : mean of x

σ_x^2 : variance

σ_x : standard deviation

$$x_1 = \mu_x - \sigma_x$$

$$x_2 = \mu_x + \sigma_x$$

$$P_{x_1 \rightarrow x_2} = 0.6827$$

x_a, x_b	$\mu_x \pm \sigma_x$	$\mu_x \pm 1.96\sigma_x$	$\mu_x \pm 2\sigma_x$
$P_{x_a \rightarrow x_b}$	0.6827	0.95	0.96

R commands for Gaussian Distribution

`prnorm(q=1, mean=2, sd=2)` # Gives C(1) on $P_{-\infty \rightarrow 1}$

Example problem

x is normal with $\mu_x = 2$
 $\sigma_x = 4$

What is the probability that a single measurement of x will be $-1 \leq x \leq 9$?

Ans: $P_{-1 \rightarrow 9}$

Z-TRANSFORMATION

Consider $P_{\mu_x \rightarrow x}$

$$\text{define } \frac{x - \mu_x}{\sigma_x} = z; dz = \frac{d(x - \mu_x)}{\sigma_x} = \frac{dx}{\sigma_x}$$

$$\text{so, } \frac{dx}{\sigma_x} = \frac{1}{\sqrt{2\pi}} \int_0^x \exp \left[-\frac{z^2}{2} \right] dz = P_{0 \rightarrow z}$$

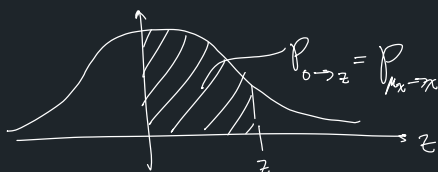


Table 4.3 gives value for this